ASK Phase

Business Problem:

Cyclistic aims to increase annual memberships because members are more profitable than casual riders. The marketing team wants to understand how usage differs between the two groups in order to design targeted conversion strategies.

Stakeholders:

- Lily Moreno - Director of Marketing
- Cyclistic Marketing Analytics Team
- Cyclistic Executive Team

Prepare Phase

Description of Data Sources Used:

The primary dataset used for this analysis is the Cyclistic historical bike trip dataset, which contains public trip level ride data made available by Motivate International Inc. under a public license. The dataset includes detailed records of individual bike share trips taken within the Cyclistic system.

The scope of the data used for this case study includes 12 months of trip data for the year 2025, which were downloaded as individual monthly CSV files. Each file contains ride level information including ride start time, ride end time, bike type, station information, and rider type classification (member or casual).

Data Storage and Organization:

After downloading the dataset, the files were initially stored locally and then imported into Google BigQuery to allow efficient processing of the large dataset. Because the files exceeded typical spreadsheet size limitations, BigQuery was used as the primary tool for storage and querying.

The dataset was initially organized as separate monthly tables, with some months split into multiple files due to file size limits. These monthly datasets were later combined into a single unified table using SQL queries with the `UNION ALL` function. This created a complete dataset representing all rides taken during the 12 month period, which allowed for full year analysis of ride behavior.

Data Credibility and Limitations:

The dataset is considered credible for this analysis because it represents real operational ride data collected by the Cyclistic bike share system. The dataset is appropriate for examining usage patterns between rider types because it includes variables such as ride timestamps, ride duration, bike type, and rider classification.

However, there are several limitations associated with the dataset. Due to data privacy protections, the dataset does not contain any personally identifiable information. As a result, rides cannot be linked to individual users, payment methods, or demographic characteristics. This prevents deeper analysis of rider demographics or individual customer behavior.

Despite these limitations, the available variables provide sufficient information to analyze behavioral differences between casual riders and annual members, which is the central business question of the case study.

How the Data Supports the Business Question:

The dataset includes key variables such as ride start time, ride end time, rider type (member or casual), and station location information, which allow for analysis of ride duration, ride frequency, seasonal trends, and weekday usage patterns.

These variables make it possible to identify differences in how casual riders and annual members use the bike share system. By analyzing these patterns, insights can be generated to support Cyclistic's marketing objective of converting casual riders into annual members.

Process Phase

Business Task:

Analyze Cyclistic bike share trip data to identify behavioral differences between annual members and casual riders. The goal is to generate insights that can help the marketing team design strategies to convert casual riders into annual members.

Data Sources Used:

The dataset used for this analysis is the Cyclistic historical bike trip dataset provided as part of the case study. The data consists of monthly ride records for the year 2025.

The original data was provided as multiple CSV files representing individual months. Due to the large dataset size, the data was imported and processed using **Google BigQuery**, which is capable of handling large datasets efficiently.

All individual monthly datasets were combined into a single table called:

```
cyclistic_2025_all
```

This table served as the base dataset for cleaning, transformation, and analysis.

Data Integrity Verification:

Before cleaning the data, several checks were performed to confirm the dataset loaded correctly and matched expectations.

Dataset Size:

Total number of rides in the dataset:

Total rides:
5,552,994

This confirms the data was successfully combined from all monthly files into a single dataset.

Duplicate Ride ID Check:

To verify that each ride record is unique, the ride_id field was checked for duplicates.

```
SELECT

ride_id,

COUNT(*)

FROM cyclistic_2025_all

GROUP BY ride_id

HAVING COUNT(*) > 1
```

No duplicate ride IDs were detected, confirming each ride represents a unique trip.

Rider Type Validation:

To verify the expected rider categories exist in the dataset, the following query was used:

```
SELECT

member_casual,

COUNT(*) AS rides

FROM cyclistic_2025_all

GROUP BY member_casual
```

Results:

Rider Type	Total Rides
Member	3,553,497
Casual	1,999,497

The dataset contains two rider types: **members** and **casual riders**, which matches the expected structure of the Cyclistic dataset.

Data Quality Checks:

Several checks were performed to identify potential data quality issues before cleaning.

Missing Station Values:

Some rides contained missing start or end station information.

```
SELECT

COUNT(*) AS total_rides,

COUNTIF(start_station_name IS NULL) AS null_start_station,

COUNTIF(end_station_name IS NULL) AS null_end_station

FROM cyclistic_2025_all
```

Results:

Metric	Value

Total Rides	5,552,994
Null Start Station	1,184,673
Null End Station	1,243,305

These null values likely occur when bikes are started or ended outside of fixed docking stations. Because station information is not required for analyzing rider behavior or ride duration, these rows were retained in the dataset.

Data Cleaning Steps:

The dataset was cleaned using SQL transformations to ensure that only valid ride records were included in the analysis.

The following filters were applied:

• Removed rows where **ended_at occurred before started_at**
• Removed rows where **timestamps were missing**
• Created calculated fields for ride duration and time based analysis

Cleaning query used:

```
CREATE OR REPLACE TABLE cleaned_trips AS

SELECT *,

TIMESTAMP_DIFF(ended_at, started_at, MINUTE) AS ride_length_minutes,

EXTRACT(DAYOFWEEK FROM started_at) AS day_of_week,

EXTRACT(MONTH FROM started_at) AS month,

FORMAT_TIMESTAMP('%A', started_at) AS weekday_name

FROM cyclistic_2025_all

WHERE

ended_at > started_at

AND started_at IS NOT NULL

AND ended_at IS NOT NULL
```

This query created a cleaned dataset called **cleaned_trips**, which was used for further analysis.

Data Transformation:

Additional variables were created to support behavioral analysis.

Variable	Description
ride_length_minutes	Duration of each ride in minutes
day_of_week	Numeric representation of the day of the week
weekday_name	Full name of the weekday
month	Month in which the ride occured

These variables allow analysis of ride behavior based on **ride duration, day of the week, and monthly trends**.

Data Validation:

After cleaning the data, validation checks were performed to ensure that the calculated values were reasonable.

Query used:

```
SELECT

MIN(ride_length_minutes),

MAX(ride_length_minutes),

AVG(ride_length_minutes)

FROM cleaned_trips
```

Results:

Metric	Value
Minimum Ride Length	1 minute
Maximum Ride Length	1439 minutes

Average Ride Length	14.42 minutes

Final Dataset:

After cleaning and transformation, the dataset was validated to ensure that all rides contained valid timestamps, reasonable ride durations, and correctly formatted variables required for behavioral analysis.

The final cleaned dataset contains **5,400,008 rides.**

The final cleaned dataset is stored in the table:

```
cleaned_trips
```

This dataset is now prepared for the **Analyze phase**, where ride behavior patterns between members and casual riders will be examined.

Analysis Phase

The analysis phase examined behavioral differences between Cyclistic annual members and casual riders using the cleaned dataset created during the Process phase. The dataset contained **5,400,008 cleaned ride records** representing trips taken during 2025.

The goal of the analysis was to identify usage patterns that distinguish casual riders from annual members in order to support marketing strategies aimed at converting casual riders into annual members.

Ride Frequency and Average Ride Duration:

The first analysis examined the total number of rides taken by each rider type and their average ride duration using the query:

```
SELECT

member_casual,

COUNT(*) AS rides,

AVG(ride_length_minutes) AS avg_ride_length

FROM `cyclistic-case-study-488620.cyclistic_trips.cleaned_trips`
```

```
GROUP BY member_casual
```

Results:

Rider Type	Total Rides	Average Ride Length
Casual	1,915,806	19.4 minutes
Member	3,484,202	11.7 minutes

Members take significantly more trips overall, indicating they use the bike share system more frequently as part of regular transportation routines.

Casual riders, however, take longer rides on average. This suggests casual riders may be using the bikes for **recreational or leisure activities**, while members likely use the service for **shorter, routine trips such as commuting or errands**.

Ride Patterns by Day of Week:

Ride frequency was then analyzed by weekday for both rider types.

Results show that **members consistently take more rides during weekdays**, particularly on Tuesday, Wednesday, Thursday, and Friday. These patterns suggest that members rely on Cyclistic bikes as part of their **daily transportation or commuting routines**.

Casual riders also take rides during weekdays, but their ride counts are comparatively lower. However, casual riders show relatively stronger usage on **weekends**, especially Saturday and Sunday.

This pattern supports the hypothesis that casual riders are more likely to use Cyclistic bikes for **recreational purposes, tourism, or occasional leisure trips**, rather than regular commuting.

```sql
SELECT
weekday_name,
member_casual,
COUNT(*) AS rides
FROM `cyclistic-case-study-488620.cyclistic_trips.cleaned_trips`
GROUP BY weekday_name, member_casual
ORDER BY rides DESC
```

Query results

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Job information | Results | Visualization | JSON | Execution details | Execution gra

Row	weekday_name	member_casual	rides
1	Thursday	member	565177
2	Tuesday	member	552529
3	Wednesday	member	540212
4	Friday	member	518551
5	Monday	member	493322
6	Saturday	member	439876
7	Saturday	casual	395595
8	Sunday	member	374535
9	Sunday	casual	316953
10	Friday	casual	306463
11	Thursday	casual	247893
12	Monday	casual	219232
13	Tuesday	casual	216926
14	Wednesday	casual	212744

Ride Duration by Day of Week:

The analysis also examined how ride duration varies by day of week for both rider types.

Results show that casual riders have longer average ride durations across all days of the week. The longest average rides occur on **weekend days**, particularly Sunday and Saturday.

Members maintain shorter and more consistent ride durations across the entire week.

This reinforces the interpretation that:

Members tend to use bikes for **short functional trips**, while casual riders tend to take **longer recreational rides**.

```sql
SELECT
weekday_name,
member_casual,
AVG(ride_length_minutes) AS avg_ride_length
FROM `cyclistic-case-study-488620.cyclistic_trips.cleaned_trips`
GROUP BY weekday_name, member_casual
ORDER BY avg_ride_length DESC
```

Query results

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Job information | Results | Visualization | JSON | Execution details | Execution graph

Row	weekday_name	member_casual	avg_ride_length
1	Sunday	casual	22.60348064224...
2	Saturday	casual	21.90298158470...
3	Monday	casual	19.20068694351...
4	Friday	casual	19.10061899805...
5	Tuesday	casual	17.11739948185...
6	Thursday	casual	16.94940559031...
7	Wednesday	casual	15.86206426503...
8	Sunday	member	12.94716648644...
9	Saturday	member	12.80640907892...
10	Friday	member	11.64245175498...
11	Tuesday	member	11.34676369928...
12	Monday	member	11.25845999164...
13	Thursday	member	11.22619993382...
14	Wednesday	member	11.12732223645...

Results per page: 50

Ride Patterns by Month:

Ride frequency was also analyzed by month to identify seasonal patterns.

The results show that ride activity increases significantly during the summer months, particularly June through September. Both rider types increase usage during this period, but members still account for the majority of total rides.

Peak ride counts occur during late summer months, which suggests that warmer weather and seasonal tourism increase overall bike usage.

Casual riders show noticeable increases during these peak months, indicating that recreational usage may rise during favorable weather conditions.

```sql
SELECT
month,
member_casual,
COUNT(*) AS rides
FROM `cyclistic-case-study-488620.cyclistic_trips.cleaned_trips`
GROUP BY month, member_casual
ORDER BY rides DESC
```

Query results

+ Create conversation Save results

Job information | Results | Visualization | JSON | Execution details | Execution graph

Row	month	member_casual	rides
1	8	member	443130
2	9	member	440954
3	7	member	430394
4	10	member	414094
5	6	member	379523
6	8	casual	323533
7	5	member	313996
8	7	casual	308446
9	6	casual	278702
10	4	member	257921
11	9	casual	254727
12	11	member	251924
13	10	casual	214380
14	3	member	208458
15	5	casual	175655

Results per page: 50 1 – 24 of

Query results

+ Create conversation Save results

Job information | Results | Visualization | JSON | Execution details | Execution graph

Row	month	member_casual	rides
10	4	member	257921
11	9	casual	254727
12	11	member	251924
13	10	casual	214380
14	3	member	208458
15	5	casual	175655
16	2	member	122097
17	1	member	112331
18	12	member	109380
19	4	casual	105260
20	11	casual	94719
21	3	casual	82864
22	12	casual	27112
23	2	casual	27003
24	1	casual	23405

Results per page: 50 1 – 24 of 24

Job history

Bike Type Usage:

Bike usage was analyzed by rider type to determine whether casual riders and members show different preferences for Cyclistic bike types.

Members accounted for the majority of rides across both bike types. For **classic bikes**, members completed **1,274,451 rides (65.61%)**, while casual riders completed **667,993 rides (34.39%)**. For **electric bikes**, members completed **2,209,751 rides (63.91%)**, compared to **1,247,813 rides (36.09%)** by casual riders.

Although members dominate overall usage, casual riders make up a slightly larger share of electric bike rides compared to classic bikes. This may indicate that casual riders are more likely to choose electric bikes due to their convenience and ease of use.

```sql
SELECT
rideable_type,
member_casual,
COUNT(*) AS rides,
ROUND(
COUNT(*) * 100.0 /
SUM(COUNT(*)) OVER(PARTITION BY rideable_type),
2
) AS percent_share
FROM `cyclistic-case-study-488620.cyclistic_trips.cleaned_trips`
GROUP BY rideable_type, member_casual
ORDER BY rideable_type;
```

Query completed

Using on-demand processing quota

Query results

+ Create conversation

Job information | Results | Visualization | JSON | Execution details | Execution graph

Row	rideable_type	member_casual	rides	percent_share
1	classic_bike	member	1274451	65.61
2	classic_bike	casual	667993	34.39
3	electric_bike	member	2209751	63.91
4	electric_bike	casual	1247813	36.09

Key Behavioral Differences Identified

The analysis revealed several clear behavioral differences between casual riders and annual members.

Members

- Take significantly more rides overall
- Ride more consistently throughout the week
- Have shorter average ride durations
- Likely use Cyclistic bikes for commuting or routine transportation

Casual Riders

- Take fewer total rides
- Have longer average ride durations
- Show stronger weekend usage patterns
- Likely use bikes for leisure or recreational purposes

Analytical Conclusion

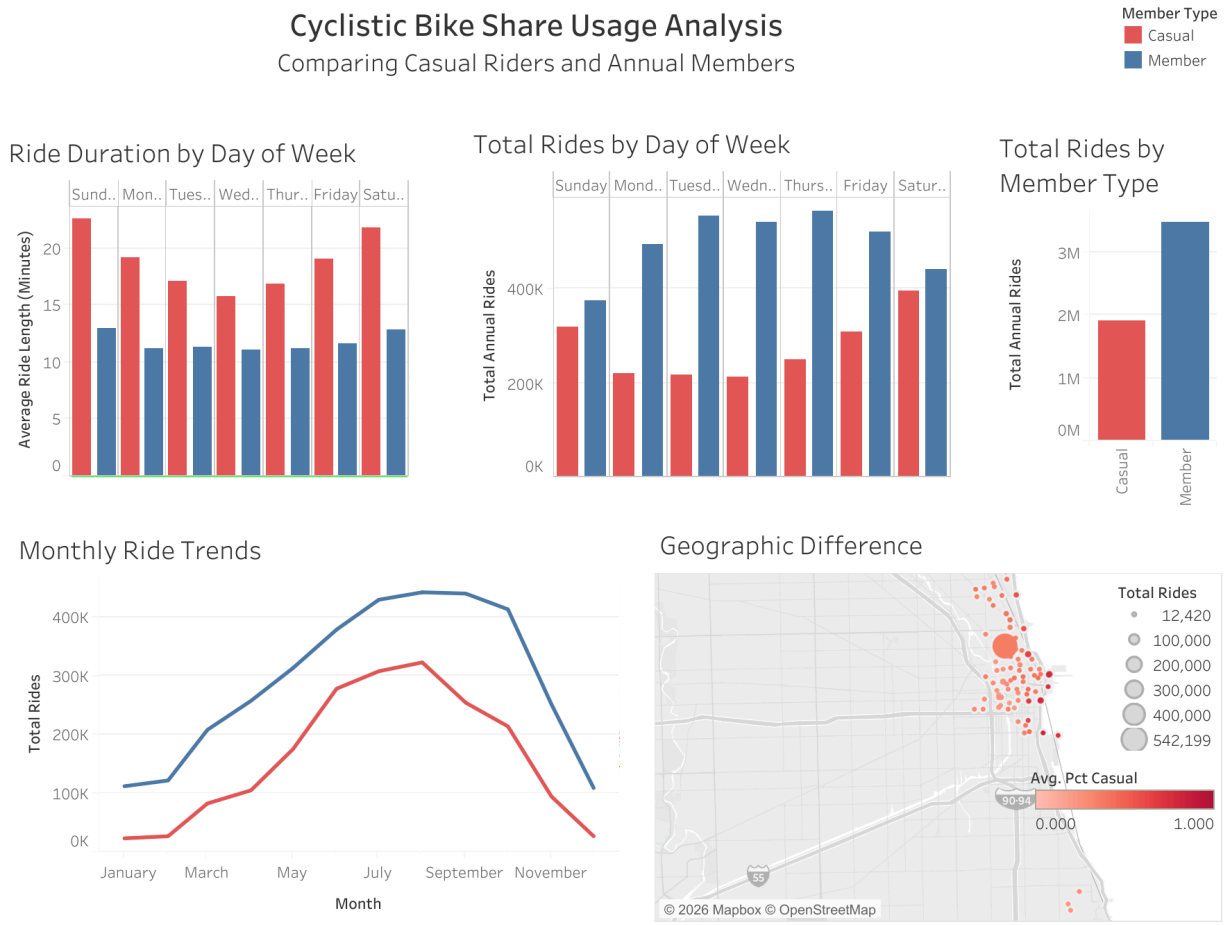
The results suggest that Cyclistic members tend to use the bike share system as part of their daily transportation routine, while casual riders appear to use the bikes more for recreational or occasional trips.

These differences suggest that marketing efforts aimed at converting casual riders into members should emphasize:

- Cost savings for frequent riders
- The convenience of unlimited rides with a membership
- Membership benefits for riders who regularly take longer leisure trips

Understanding these usage patterns provides valuable insight for the Cyclistic marketing team when designing targeted campaigns to convert casual riders into annual members.

Share Phase



https://public.tableau.com/views/CyclisticBikeShareAnalysis_17725693546420/Dashboard1?:language=en-US&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link

https://console.cloud.google.com/bigquery?ws=!1m4!1m3!3m2!1scyclistic-case-study-488620!2scyclistic_trips

Act Phase

Top Three Recommendations:

1. Promote membership benefits for frequent riders

Cyclistic should highlight the cost savings and convenience of annual memberships compared to single ride or daily passes. Since members take significantly more rides and use the service more consistently throughout the week, marketing campaigns should emphasize unlimited rides, long term savings, and convenience for riders who frequently use the bikes for transportation.

2. Target casual riders during peak recreational periods

Casual riders show higher usage on weekends and during the summer months. Cyclistic could launch seasonal marketing campaigns or limited time membership discounts during these peak recreational periods to encourage casual riders to convert to annual memberships.

3. Focus marketing efforts on recreational riders transitioning to regular use

Because casual riders tend to take longer rides, Cyclistic could promote memberships as a way to support regular recreational cycling. Messaging could focus on how an annual membership allows riders to explore the city more often without worrying about individual ride costs.

Next Steps for the Business:

The marketing team could use these insights to design targeted digital campaigns aimed at casual riders who already demonstrate high engagement with the bike share system. These campaigns could be distributed through mobile apps, email marketing, or social media to promote the benefits of converting to an annual membership.

An additional analysis that could strengthen these findings would be examining **ride start times throughout the day** to determine whether members and casual riders use the bike share system at different times. For example, members may show higher usage during traditional commuting hours in the morning and evening, while casual riders may ride more frequently in the afternoon or evening for leisure. Understanding these time based patterns could help Cyclistic better target marketing campaigns during specific periods when casual riders are most active.

Another potential area of analysis would be **ride distance and route patterns**. By examining how far riders travel and which stations are commonly used together, Cyclistic could identify popular recreational routes or commuter corridors. This insight could help the company place targeted advertisements or promotional messaging at stations with high casual rider traffic.

Finally, future analysis could incorporate **weather data**, such as temperature and precipitation, to determine how environmental conditions influence rider behavior. Understanding how weather impacts ride frequency could help Cyclistic plan seasonal promotions or marketing campaigns during periods when casual ridership is most likely to increase.